SG No. 219 T_{d}^{5} $F\bar{4}3c$ [cubic]

* plus set: $+[0,\,0,\,0]$, $+\left[0,\,\frac{1}{2},\,\frac{1}{2}\right]$, $+\left[\frac{1}{2},\,0,\,\frac{1}{2}\right]$, $+\left[\frac{1}{2},\,\frac{1}{2},\,0\right]$

Table 1: Wyckoff site: `8a`, site symmetry: `23.`

No.	position	mapping
1	$[0,\,0,\,0]$	`[1,2,3,4,5,6,7,8,9,10,11,12]`
2	$\left[\frac{1}{2},\,\frac{1}{2},\,\frac{1}{2}\right]$	`[13,14,15,16,17,18,19,20,21,22,23,24]`

Table 2: Wyckoff site: `8b`, site symmetry: `23.`

No.	position	mapping
1	$\left[\frac{1}{4},\,\frac{1}{4},\,\frac{1}{4}\right]$	`[1,2,3,4,5,6,7,8,9,10,11,12]`
2	$\left[\frac{3}{4},\,\frac{3}{4},\,\frac{3}{4}\right]$	`[13,14,15,16,17,18,19,20,21,22,23,24]`

Table 3: Wyckoff site: `24c`, site symmetry: `-4..`

No.	position	mapping
1	$\left[0,\,\frac{1}{4},\,\frac{1}{4}\right]$	`[1,4,18,19]`
2	$\left[0,\,\frac{3}{4},\,\frac{1}{4}\right]$	`[2,3,17,20]`
3	$\left[\frac{1}{4},\,0,\,\frac{1}{4}\right]$	`[5,8,14,15]`
4	$\left[\frac{1}{4},\,0,\,\frac{3}{4}\right]$	`[6,7,13,16]`
5	$\left[\frac{1}{4},\,\frac{1}{4},\,0\right]$	`[9,12,22,23]`
6	$\left[\frac{3}{4},\,\frac{1}{4},\,0\right]$	`[10,11,21,24]`

Table 4: Wyckoff site: `24d`, site symmetry: `-4..`

No.	position	mapping
1	$\left[\frac{1}{4},\,0,\,0\right]$	`[1,4,18,19]`
2	$\left[\frac{3}{4},\,0,\,0\right]$	`[2,3,17,20]`
3	$\left[0,\,\frac{1}{4},\,0\right]$	`[5,8,14,15]`
4	$\left[0,\,\frac{3}{4},\,0\right]$	`[6,7,13,16]`
5	$\left[0,\,0,\,\frac{1}{4}\right]$	`[9,12,22,23]`
6	$\left[0,\,0,\,\frac{3}{4}\right]$	`[10,11,21,24]`

Table 5: Wyckoff site: `32e`, site symmetry: `.3.`

No.	position	mapping
1	$[x,\,x,\,x]$	`[1,5,9]`
2	$[-x,\,-x,\,x]$	`[2,7,12]`
3	$[-x,\,x,\,-x]$	`[3,8,10]`
4	$[x,\,-x,\,-x]$	`[4,6,11]`
5	$\left[x+\frac{1}{2},\,x+\frac{1}{2},\,x+\frac{1}{2}\right]$	`[13,17,21]`
6	$\left[\frac{1}{2}-x,\,\frac{1}{2}-x,\,x+\frac{1}{2}\right]$	`[14,19,24]`
7	$\left[x+\frac{1}{2},\,\frac{1}{2}-x,\,\frac{1}{2}-x\right]$	`[15,20,22]`

continued ...

Table 5

No.	position	mapping
8	$\left[\frac{1}{2}-x,\, x+\frac{1}{2},\, \frac{1}{2}-x\right]$	[16,18,23]

Table 6: Wyckoff site: 48f, site symmetry: 2..

No.	position	mapping
1	$[x,\, 0,\, 0]$	[1,4]
2	$[-x,\, 0,\, 0]$	[2,3]
3	$[0,\, x,\, 0]$	[5,8]
4	$[0,\, -x,\, 0]$	[6,7]
5	$[0,\, 0,\, x]$	[9,12]
6	$[0,\, 0,\, -x]$	[10,11]
7	$\left[\frac{1}{2},\, x+\frac{1}{2},\, \frac{1}{2}\right]$	[13,16]
8	$\left[\frac{1}{2},\, \frac{1}{2}-x,\, \frac{1}{2}\right]$	[14,15]
9	$\left[x+\frac{1}{2},\, \frac{1}{2},\, \frac{1}{2}\right]$	[17,20]
10	$\left[\frac{1}{2}-x,\, \frac{1}{2},\, \frac{1}{2}\right]$	[18,19]
11	$\left[\frac{1}{2},\, \frac{1}{2},\, x+\frac{1}{2}\right]$	[21,24]
12	$\left[\frac{1}{2},\, \frac{1}{2},\, \frac{1}{2}-x\right]$	[22,23]

Table 7: Wyckoff site: 48g, site symmetry: 2..

No.	position	mapping
1	$\left[x,\, \frac{1}{4},\, \frac{1}{4}\right]$	[1,4]
2	$\left[-x,\, \frac{3}{4},\, \frac{1}{4}\right]$	[2,3]
3	$\left[\frac{1}{4},\, x,\, \frac{1}{4}\right]$	[5,8]
4	$\left[\frac{1}{4},\, -x,\, \frac{3}{4}\right]$	[6,7]
5	$\left[\frac{1}{4},\, \frac{1}{4},\, x\right]$	[9,12]
6	$\left[\frac{3}{4},\, \frac{1}{4},\, -x\right]$	[10,11]
7	$\left[\frac{3}{4},\, x+\frac{1}{2},\, \frac{3}{4}\right]$	[13,16]
8	$\left[\frac{1}{4},\, \frac{1}{2}-x,\, \frac{3}{4}\right]$	[14,15]
9	$\left[x+\frac{1}{2},\, \frac{3}{4},\, \frac{3}{4}\right]$	[17,20]
10	$\left[\frac{1}{2}-x,\, \frac{3}{4},\, \frac{1}{4}\right]$	[18,19]
11	$\left[\frac{3}{4},\, \frac{3}{4},\, x+\frac{1}{2}\right]$	[21,24]
12	$\left[\frac{3}{4},\, \frac{1}{4},\, \frac{1}{2}-x\right]$	[22,23]

Table 8: Wyckoff site: 96h, site symmetry: 1

No.	position	mapping
1	$[x,\, y,\, z]$	[1]
2	$[-x,\, -y,\, z]$	[2]
3	$[-x,\, y,\, -z]$	[3]
4	$[x,\, -y,\, -z]$	[4]
5	$[z,\, x,\, y]$	[5]
6	$[z,\, -x,\, -y]$	[6]

continued ...

Table 8

No.	position	mapping
7	$[-z,\, -x,\, y]$	[7]
8	$[-z,\, x,\, -y]$	[8]
9	$[y,\, z,\, x]$	[9]
10	$[-y,\, z,\, -x]$	[10]
11	$[y,\, -z,\, -x]$	[11]
12	$[-y,\, -z,\, x]$	[12]
13	$\left[y+\frac{1}{2},\, x+\frac{1}{2},\, z+\frac{1}{2}\right]$	[13]
14	$\left[\frac{1}{2}-y,\, \frac{1}{2}-x,\, z+\frac{1}{2}\right]$	[14]
15	$\left[y+\frac{1}{2},\, \frac{1}{2}-x,\, \frac{1}{2}-z\right]$	[15]
16	$\left[\frac{1}{2}-y,\, x+\frac{1}{2},\, \frac{1}{2}-z\right]$	[16]
17	$\left[x+\frac{1}{2},\, z+\frac{1}{2},\, y+\frac{1}{2}\right]$	[17]
18	$\left[\frac{1}{2}-x,\, z+\frac{1}{2},\, \frac{1}{2}-y\right]$	[18]
19	$\left[\frac{1}{2}-x,\, \frac{1}{2}-z,\, y+\frac{1}{2}\right]$	[19]
20	$\left[x+\frac{1}{2},\, \frac{1}{2}-z,\, \frac{1}{2}-y\right]$	[20]
21	$\left[z+\frac{1}{2},\, y+\frac{1}{2},\, x+\frac{1}{2}\right]$	[21]
22	$\left[z+\frac{1}{2},\, \frac{1}{2}-y,\, \frac{1}{2}-x\right]$	[22]
23	$\left[\frac{1}{2}-z,\, y+\frac{1}{2},\, \frac{1}{2}-x\right]$	[23]
24	$\left[\frac{1}{2}-z,\, \frac{1}{2}-y,\, x+\frac{1}{2}\right]$	[24]